

Soln: Conclusion: $q \rightarrow \neg r$

<u>Steps</u>	<u>Reason</u>
1. $p \vee \neg q$	Premise
2. $r \rightarrow \neg r$	Premise
3. $\neg p \vee \neg r$	Conditional-disjunction equivalence on 2
4. $\neg p \vee \neg r$	Commutative law on 3
5. $\neg q \vee \neg r$	Resolution on 1, 4
6. $q \rightarrow \neg r$	Conditional-disjunction equivalence on 5

b) That not many students like logic, if mathematics is not easy.

Soln: Conclusion: $\neg r \rightarrow \neg q$

<u>Steps</u>	<u>Reason</u>
1. $p \vee \neg q$	Premise
2. $r \rightarrow \neg r$	Premise
3. $\neg r \rightarrow \neg q$	Conditional-disjunction equivalence on 1
4. $r \rightarrow \neg q$	Hypothetical syllogism on 2, 3

$$\begin{aligned}
 \text{Clause } C_1: & r \rightarrow \neg q \equiv \neg r \vee \neg q \\
 \text{Clause } C_2: & \neg(\neg r \rightarrow \neg q) \equiv \neg(\neg(\neg r) \vee \neg q) \quad (\text{Conditional-disjunction equivalence}) \\
 & \equiv \neg(r \vee \neg q) \quad (\text{Double-negation law}) \\
 & \equiv \neg r \wedge \neg(\neg q) \quad (\text{De Morgan's law}) \equiv \neg r \wedge q \quad (\text{Double negation law}) \\
 \text{Clause } C_3: & q
 \end{aligned}$$

By Resolution on C_1, C_3 we get $C_4: \neg r$
 We cannot reach a contradiction from C_2 and C_4 .
 $\therefore$ The conclusion is not valid.